

Exact Owner Subtraction without a State Enumeration

Analytic ledger replacement, Packet F

Abstract

The executable exact ledger checks a 9×7 truth table when the accepted lower, staircase, and optical regions are subtracted from the Round-19 residual. This note replaces all sixty-three state checks and the five additional interface checks by one set-theoretic calculation. Every open and closed face is retained explicitly.

1 Sets and inherited owners

Let

$$0 < \rho_* < \rho_0 < \rho_c < \rho_{\text{opt}} < \frac{7}{8}, \quad \rho_{\text{opt}} := \frac{39}{50},$$

and let $U(\rho)$, $k_6(\rho)$, and $k_{11}(\rho)$ be the upper owner and the two staircase walls from the main proof. Put

$$D_{\text{low}} := \left\{ \rho_* < \rho \leq \rho_0, (2\rho)^{-1} < K < U(\rho) \right\} \cup \left\{ \rho_0 < \rho < \rho_c, d < K < U(\rho) \right\}, \quad (1)$$

$$D_{\text{high}} := \left\{ \rho_c \leq \rho < \frac{7}{8}, k_6(\rho) < K < U(\rho) \right\}, \quad (2)$$

where $d = (2\rho_0)^{-1}$, and define

$$D_{19} := D_{\text{low}} \dot{\cup} D_{\text{high}}. \quad (3)$$

The dot is justified by the disjoint ratio ranges.

The already-proved inclusive owners used in the subtraction are:

$$L_{\text{own}} := D_{\text{low}}, \quad (4)$$

$$S_{\text{own}} := \left\{ \rho_c \leq \rho < \frac{7}{8}, k_6(\rho) < K \leq k_{11}(\rho) \right\}, \quad (5)$$

$$O_{\text{own}} := \left\{ \rho_{\text{opt}} \leq \rho < \frac{7}{8}, K \geq 0 \right\}. \quad (6)$$

The lower owner includes the complete $\rho = \rho_0$ fibre. The staircase includes its upper wall $K = k_{11}(\rho)$. The optical owner includes its lower ratio face $\rho = \rho_{\text{opt}}$. In contrast, D_{19} already excludes $K = U(\rho)$.

2 The subtraction

Proposition 1 (Exact residual identity). *One has*

$$\boxed{D_{19} \setminus (L_{\text{own}} \cup S_{\text{own}} \cup O_{\text{own}}) = D_{20}}, \quad (7)$$

where

$$D_{20} := \{\rho_c \leq \rho < \rho_{\text{opt}}, \ k_{11}(\rho) < K < U(\rho)\}. \quad (8)$$

Proof. Since $L_{\text{own}} = D_{\text{low}}$ and the union in (3) is disjoint,

$$D_{19} \setminus L_{\text{own}} = D_{\text{high}}. \quad (9)$$

Inside D_{high} , the lower frequency inequality is strict: $k_6(\rho) < K$. Removing the inclusive staircase (5) therefore gives exactly

$$D_{\text{high}} \setminus S_{\text{own}} = \left\{ \rho_c \leq \rho < \frac{7}{8}, \ k_{11}(\rho) < K < U(\rho) \right\}. \quad (10)$$

There is no missing equality case: $K = k_{11}(\rho)$ belongs to S_{own} , while $K = U(\rho)$ never belonged to D_{19} .

Finally, subtracting the inclusive ratio owner (6) changes $\rho < 7/8$ in (10) to $\rho < \rho_{\text{opt}}$. Its equality face is removed because $\rho = \rho_{\text{opt}}$ belongs to O_{own} . The face $\rho = \rho_c$ remains, since it belongs neither to D_{low} nor to O_{own} . The result is precisely (8). $\square$

3 The five delicate interfaces

For clarity, the five extra assertions formerly checked separately by the executable ledger are consequences of the displayed sets:

1. At $\rho = \rho_0$, the identity $(2\rho_0)^{-1} = d$ makes the prospective second lower fibre empty; the first set in (1) owns the full equality fibre, and L_{own} removes it exactly once.
2. The face $\rho = \rho_c$ belongs to D_{high} . It is not an optical face, so points with $k_{11}(\rho_c) < K < U(\rho_c)$ remain in D_{20} .
3. The face $\rho = \rho_{\text{opt}} = 39/50$ belongs to O_{own} and is therefore absent from D_{20} .
4. The frequency face $K = k_{11}(\rho)$ belongs to S_{own} , whereas $K = U(\rho)$ is excluded already by the strict inequality in D_{19} .
5. If $\rho_c < \rho < \rho_{\text{opt}}$ and $k_{11}(\rho) < K < U(\rho)$, the point is in D_{high} but in none of $L_{\text{own}}, S_{\text{own}}, O_{\text{own}}$; hence it survives, and every surviving point has exactly these properties.

4 Final compact/tail split

Define

$$C_{21} := D_{20} \cap \{K \leq 200\}, \quad T_{21} := D_{20} \cap \{K > 200\}. \quad (11)$$

The elementary partition $(0, \infty) = \{K \leq 200\} \dot{\cup} \{K > 200\}$ gives

$$D_{20} = C_{21} \dot{\cup} T_{21}, \quad D_{20} \setminus (C_{21} \cup T_{21}) = \emptyset. \quad (12)$$

Thus $K = 200$ is assigned uniquely to the compact part, regardless of whether $U(\rho)$ lies below, at, or above 200.

Ledger rows replaced. Proposition 1 replaces all $9 \cdot 7 = 63$ state evaluations in the executable family `verify_exact_subtraction`. The five items above replace its five separately asserted interface predicates. Equation (12) supplies the subsequent compact/tail subtraction without a classifier or executable truth table.